

Survival analysis with actuarial applications (I)

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MESIO

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References

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Introduction to survival analysis

Introduction

The survival analysis has different names:

- Survival analysis
- Lifetime data analysis
- Duration analysis
- Transition analysis
- Failure time analysis

Introduction to survival analysis

Introduction

Notation:

- T is the random variable (r.a.) time until the event of interest ε and we assume that it has a continuous distribution (although in most of the situations it is discretized).
- The distribution of the r.v. T is characterized by using 6 functions:
 - Survival function: $S(t)$
 - Distribution function: $F(t)$ (Cumulative distribution function, cdf)
 - Density function: $f(t)$ (Probability density function, pdf)
 - Hazard function: $\lambda(t)$
 - Cumulative hazard function: $\Lambda(t)$
 - Mean residual life: $vr_m(t)$
- If we know any of these 6 functions, the other 5 can be obtained.
- Each of them illustrates different characteristics of the distribution.

Notation

Distribution function and density function

- The distribution function $F(t)$ and the density function $f(t)$ are very well known.

$$F(t) = P(T \leq t) = \int_0^t f(u) \, du \quad (\text{continuous})$$

$$F(t) = P(T \leq t) = \sum_{j=0}^t p(j) \quad (\text{discrete})$$

- We analyse, therefore, the rest of functions more in detail.
- The description of functions is done in the context of mortality tables.

Notation

Survival function $S(t)$

- Is the probability that an individual survives more than t years.

$$S(t) = P(T > t) = 1 - F(t) = \int_t^{+\infty} f(u) \, du \quad (\text{continuous})$$

$$S(t) = P(T > t) = \sum_{j=t+1}^{+\infty} p(j) = \sum_j p(j) 1[(j - t) > 0] \quad (\text{discrete})$$

where $1[q]$ takes the value 1 if q is true.

- Basic properties:
 - $S(0) = 1$ and $S(+\infty) = 0$
 - $S(t)$ is monotonically decreasing
 - $f(t) = \frac{-dS(t)}{dt} = \frac{dF(t)}{dt}$

Notation

Hazard function $\lambda(t)$

- Is the probability of death in a very short time interval, under the condition that the person survives until the beginning of this interval.

$$\lambda(t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} P(t \leq T < t + \Delta t \mid T \geq t)$$

- $\lambda(t) \Delta t$: is the probability that the event ε occurs for an individual aged t in the next instant. Is the proportion of individuals that fail in the instant t among those who did not fail before.
- It is also known as:
 - Conditional failure time (in the industry)
 - Mortality force (in demography)
 - Intensity function (in stocastical processes)
 - Failure rate (in epidemiology)
 - Equivalent to one minus the inverse of the Mills ratio (in economy)

Notation

Hazard function $\lambda(t)$

Basic properties:

1. $\lambda(t)$ is non-negative.
3. If T is continuous:

$$\lambda(t) = \frac{f(t)}{S(t)} = -\frac{d}{dt} \ln[S(t)]$$

$$f(t) = \lambda(t) S(t)$$

Proof:

$$\lambda(t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} P(t \leq T < t + \Delta t \mid T \geq t) = \lim_{\Delta t \rightarrow 0} \frac{1}{\Delta t} \frac{P(t \leq T < t + \Delta t)}{P(T \geq t)} = \frac{f(t)}{S(t)} = -\frac{d}{dt} \ln[S(t)]$$

$$\text{Moreover } f(t) = [1 - S(t)]' = \frac{d}{dt} [1 - S(t)] = -\frac{d}{dt} S(t) \text{ and we get } \lambda(t) = \frac{-\frac{d}{dt} S(t)}{S(t)} = -\frac{d}{dt} \ln[S(t)]$$

Notation

Hazard function $\lambda(t)$

4. If T is discrete and takes values $t_1 < t_2 < \dots$, the hazard function is defined as:

$$\lambda(t_j) = \frac{P(t_j)}{S(t_{j-1})} = \frac{S(t_{j-1}) - S(t_j)}{S(t_{j-1})} = 1 - \frac{S(t_j)}{S(t_{j-1})}$$

5. If T is discrete and takes values $t_1 < t_2 < \dots$, it can be easily proved that the survival function can be represented as the product of 1 minus the hazard function:

$$S(t) = \prod_{t_j \leq t} \frac{S(t_j)}{S(t_{j-1})} = \prod_{t_j \leq t} [1 - \lambda(t_j)]$$

Proof:

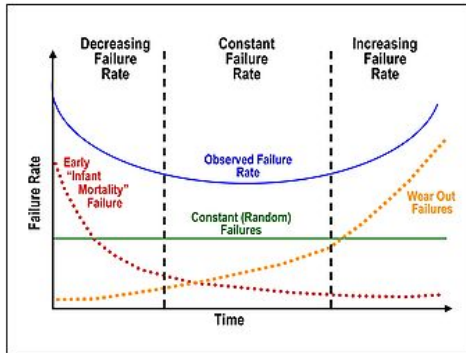
$$\begin{aligned} S(t_j) &= P(T > t_j) = P(T > t_j, T > t_{j-1}) = P(T > t_j | T > t_{j-1}) P(T > t_{j-1}) = \\ &= [1 - P(T \leq t_j | T > t_{j-1})] S(t_{j-1}) = \left[1 - \frac{P(t_j)}{S(t_{j-1})} \right] S(t_{j-1}) = [1 - \lambda(t_j)] S(t_{j-1}) \end{aligned}$$

Recursively, we obtain the former product.

Notation

Hazard function $\lambda(t)$

- The hazard function has different forms: increasing, decreasing, constant, U shape



Notation

Cumulative hazard function $\Lambda(t)$

$$\Lambda(t) = \int_0^t \lambda(u) \, du \quad (\text{It does not have any intuitive interpretation})$$

- Basic properties:

1. $\Lambda(t)$ is non negative
2. $\Lambda(t)$ is monotonically increasing
3. $\Lambda(t) = -\ln[S(t)] \implies S(t) = \exp[-\Lambda(t)]$
4. If T is discrete and takes values $t_1 < t_2 < \dots$, then $\Lambda(t) = \sum_{t_j \leq t} \lambda(t_j)$.

Notation

Mean residual life $\text{vr}_m(t)$

- Is the expectation of the remaining life time:

$$\text{vr}_m(t) = E[T - t | T > t]$$

- Basic properties:

1. $E(T) = \text{vr}_m(0) = \int_0^{+\infty} S(u) du$ is non negative.
2. If T is continuous

$$\text{vr}_m(t) = \frac{\int_t^{+\infty} S(u) du}{S(t)}$$

Censoring and Truncation

Definition

- Both censoring and truncation mean incomplete information.
- Censoring occurs when, for some observations, the only information that we have is that, in a particular time interval, the event of interest has not occur.
- Truncation occurs when some individuals (who correspond to a part of the distribution) have not been selected as part of the sample.
- The type of censoring depends on how is the time interval in which the event does not occur.
- Depending on how the follow up is carried out in the study, there are different types of censoring. The so-called right censoring is probably the one that we find the most in actuarial applications.

Censoring and truncation

Informative and non informative censoring

NON INFORMATIVE CENSORING

- The censoring time does not provide more information about the future survival of the individual than the one that we would have if the individual would have continued in the study.
- The censored individuals should have the same risk of future death than those who are not censored and are still alive.
- In statistical terms **non informative censoring** implies that the survival time and the censoring time are independent.

INFORMATIVE CENSORING

- When the survival time and the censoring time are dependent then special modelizations are required.

Right censoring

Type I right censoring

The censoring time can be different for each individual.

Let $C_1, \dots, C_m$ the censoring times, we measure:

$$Y_i = \min(T_i, C_j)$$

$$\delta_i = \begin{cases} 1 & \text{if } T_i \leq C_j \\ 0 & \text{if } T_i > C_j \end{cases} \quad \forall i = 1 \dots n, \quad j = 1, \dots, m$$

Non parametric estimation

Kaplan Meier estimator

- We assume the following:
- Let $(Y_1, \delta_1), \dots, (Y_n, \delta_n)$ be a sample of n observations, where:

$$Y_i = \min(T_i, C_i)$$

$$\delta_i = \begin{cases} 1 & \text{if } T_i \leq C_i \\ 0 & \text{if } T_i > C_i \end{cases}$$

We assume non informative censoring.

Non parametric estimation

Kaplan Meier estimator

- Let $t_1 < t_2 < \dots < t_c$ the different moments in time where an event can occur.
- Let d_j be the number of individuals who die in the moment t_j
- We define R_j as the **risk set** in the moment t_j , that is to say, the number of individuals alive right before t_j . $R_j = \sum_{i=1}^n I(Y_{(i)} \geq t_j)$
- The Kaplan Meier estimator of the survival function is:

$$\hat{S}_{KM}(t) = \begin{cases} 1 & \text{si } t < t_1 \\ \prod_{t_j \leq t} \left(1 - \frac{d_j}{R_j}\right) & \text{si } t \geq t_1 \end{cases}$$

- The cumulative hazard function is obtained as $\hat{\Lambda}_{KM}(t) = -\ln(\hat{S}_{KM}(t))$

Non parametric estimation

Variance of the Kaplan Meier estimator

- The variance of the Kaplan Meier estimator is asymptotically approximated by using the Greenwood formula, which is equivalent to

$$V_G \left(\widehat{S}_{KM}(t) \right) = \left[\hat{S}_{KM}(t) \right]^2 \sum_{t_j \leq t} \frac{d_j}{R_j (R_j - d_j)}$$

- An alternative to the Greenwood formula is the Aalen and Johansen (1978) formula, which is equivalent to

$$V_{AJ} \left(\widehat{S}_{KM}(t) \right) = \left[\hat{S}_{KM}(t) \right]^2 \sum_{t_j \leq t} \frac{d_j}{R_j^2}$$

Non parametric estimation

Nelson Aalen estimator

- First, it was proposed by Nelson (1972, Technometrics) and later on it was rediscovered by Aalen (1978, Annals of Statistics).
- The estimator is defined as follows:

$$\hat{\Lambda}_{NA}(t) = \begin{cases} 0 & \text{if } t < t_1 \\ \sum_{t_j \leq t} \frac{d_j}{R_j} & \text{if } t \geq t_1 \end{cases}$$

- The survival function can be obtained as: $\hat{S}_{NA}(t) = \exp\left(-\hat{\Lambda}_{NA}(t)\right)$.

Non parametric estimation

Variance of the Nelson Aalen estimator

- The variance of the Nelson Aalen was proposed by Aalen and Johansen in 1978:

$$V_{AJ} \left(\widehat{\hat{\Lambda}_{NA}}(t) \right) = \sum_{t_j \leq t} \frac{d_j}{R_j^2}$$

- The variance of the survival function equals:

$$V_{AJ} \left(\widehat{\hat{S}_{NA}}(t) \right) = \left[\hat{S}_{NA}(t) \right]^2 \sum_{t_j \leq t} \frac{d_j}{R_j^2}$$

This expression it is also used as an alternative to estimate the variance of the Kaplan Meier estimator.

Non parametric estimation

Log-rank and Wilcoxon tests

- ▶ Consider separately each death time in two groups of survival data, labelled Group I and Group II.
- ▶ Let $t_{(1)} < t_{(2)} < \dots < t_{(r)}$ be the r distinct failure times across the two groups.
- ▶ At time $t_{(j)}$, d_{1j} individuals in Group I and d_{2j} individuals in Group II die, for $j = 1, 2, \dots, r$, and $d_j = d_{1j} + d_{2j}$
- ▶ Suppose further that there are $n_{1j}(n_{2j})$ individuals at risk of death in the first group (second group) just before time $t_{(j)}$.

Non parametric estimation

Log-rank and Wilcoxon tests

- ▶ We regard d_{1j} as hypergeometric random variable, which can take any value in the range from 0 to the minimum of d_j and n_{1j} .
- ▶ The probability that the random variable associated with the number of deaths in the first group takes the value d_{1j} is

$$\frac{\binom{d_j}{d_{1j}} \binom{n_j - d_j}{n_{1j} - d_{1j}}}{\binom{n_j}{n_{1j}}}$$

Non parametric estimation

Log-rank and Wilcoxon tests

- ▶ The mean of the hypergeometric random variable d_{1j} is given by

$$e_{1j} = n_{1j}d_j/n_j,$$

so that e_{1j} is the expected number of individuals who die at time $t_{(j)}$ in Group I.

- ▶ The variance of d_{1j} is given by

$$v_{1j} = \frac{n_{1j}n_{2j}d_j(n_j - d_j)}{n_j^2(n_j - 1)}$$

Non parametric estimation

Log-rank and Wilcoxon tests

- ▶ The statistic is given by

$$U_L = \sum_{j=1}^r (d_{1j} - e_{1j}).$$

- ▶ The variance of U_L is $V_L = Var(U_L) = \sum_{j=1}^r v_{1j}$.
- ▶ We obtain the log-rank test statistic as

$$W_L = \frac{U_L^2}{V_L} \sim \chi_1^2.$$

- ▶ The p-value for W_L can be obtained using the χ^2 distribution with one degrees of freedom (d.f.) (i.e. $p = P(\chi^2(1) \geq W_L)$).

Non parametric estimation

Log-rank and Wilcoxon tests

- ▶ The Wilcoxon test is based on the statistic

$$U_W = \sum_{j=1}^r n_j (d_{1j} - e_{1j}).$$

- ▶ In the Wilcoxon test, each difference $d_{1j} - e_{1j}$ is weighted by n_j , the total number of individuals at risk at time $t_{(j)}$.
- ▶ The variance of U_W is given by $V_W = \sum_{j=1}^r n_j^2 v_{1j}$.
- ▶ The Wilcoxon test statistic $W_W = U_W^2 / V_W$ has a χ^2 distribution on 1 d.f. when the null hypothesis is true.

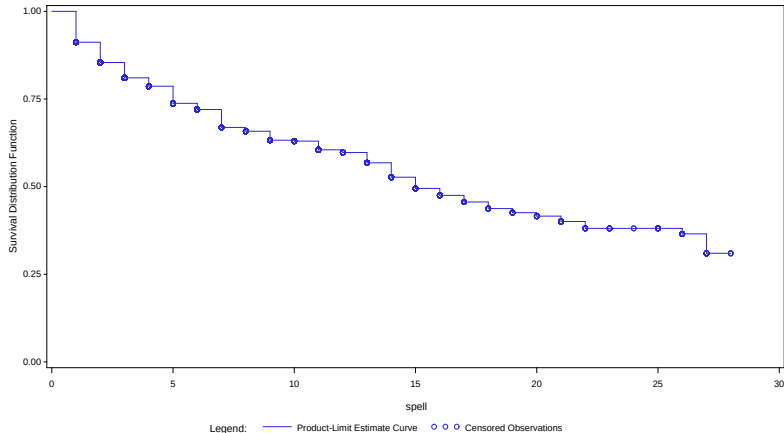
Non parametric estimation

Examples with SAS

```
*Non parametric models;  
*Graphs of the survival function;  
proc lifetest data=data plots=(survival) notable;  
time spell*event(0);  
run;  
  
*Survival curves for each group;  
proc lifetest data=data plots=(survival) notable;  
time spell*event(0);  
strata ui;  
run;
```

Non parametric estimation

Examples with SAS



Non parametric estimation

Examples with SAS

```

The LIFETEST Procedure

Summary of the Number of Censored and Uncensored Values

Stratum      ui      Total  Failed   Censored   Percent
              1      1495    576      919      61.47
              2      1848    497     1351      73.11
-----
Total              3343    1073     2270      67.90

Testing Homogeneity of Survival Curves for spell over Strata

Rank Statistics
              ui      Log-Rank      Wilcoxon
              0      228.79      671023
              1     -228.79     -671023

Covariance Matrix for the Log-Rank Statistics
              ui      0      1
              0      211.365     -211.365
              1     -211.365      211.365

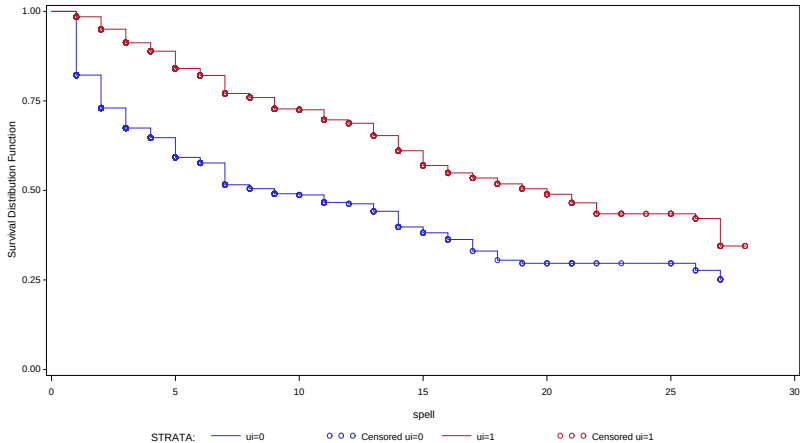
Covariance Matrix for the Wilcoxon Statistics
              ui      0      1
              0      1.3063E9     -1.306E9
              1     -1.306E9      1.3063E9

Test of Equality over Strata
              Pr >
Test      Chi-Square      DF      Chi-Square
Log-Rank      247.6590      1      <.0001
Wilcoxon      344.6992      1      <.0001
-2Log(LR)      275.3243      1      <.0001

```

Non parametric estimation

Examples with SAS



Non parametric estimation

Examples with R

```
install.packages("survival")
library(survival)
mydata<- read.csv("D:survival_unemployment.csv")
attach(mydata)

# Variable definitions
time <- spell
event <- event
X <- cbind(logwage, ui, age)
group <- ui

# Kaplan Meier
kmsurvival <- survfit(Surv(time,event) ~ 1)
summary(kmsurvival)
plot(kmsurvival, xlab="Time", ylab="Survival Probability")

# Kaplan Meier by groups
kmsurvival1 <- survfit(Surv(time, event) ~ group)
summary(kmsurvival1)
plot(kmsurvival1, xlab="Time", ylab="Survival Probability")

# Nelson Aalen
nasurvival <- survfit(coxph(Surv(time,event)~1), type="aalen")
summary(nasurvival)
plot(nasurvival, xlab="Time", ylab="Survival Probability")
```

Non parametric estimation

Examples with R

```
> kmsurvival <- survfit(Surv(time,event) ~ 1)
> summary(kmsurvival)
```

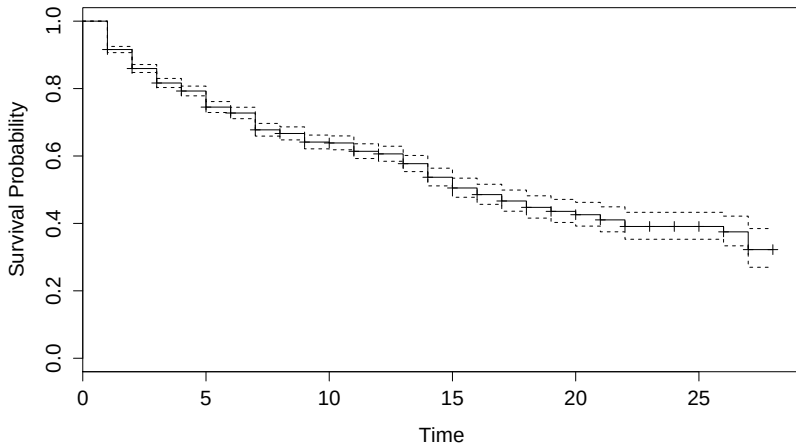
```
Call: survfit(formula = Surv(time, event) ~ 1)
```

time	n.risk	n.event	survival	std.err	lower 95 % CI	upper 95 % CI
1	3343	294	0.912	0.00490	0.903	0.922
2	2803	178	0.854	0.00622	0.842	0.866
3	2321	119	0.810	0.00708	0.797	0.824
4	1897	56	0.786	0.00756	0.772	0.801
5	1676	104	0.738	0.00847	0.721	0.754
6	1339	32	0.720	0.00882	0.703	0.737
7	1196	85	0.669	0.00979	0.650	0.688
8	933	15	0.658	0.01001	0.639	0.678
9	848	33	0.632	0.01057	0.612	0.654
10	717	3	0.630	0.01064	0.609	0.651
11	659	26	0.605	0.01128	0.583	0.627
12	556	7	0.597	0.01150	0.575	0.620
13	509	25	0.568	0.01234	0.544	0.593
14	415	30	0.527	0.01353	0.501	0.554
15	311	19	0.495	0.01458	0.467	0.524
16	252	10	0.475	0.01527	0.446	0.506
17	201	8	0.456	0.01606	0.426	0.489
18	169	7	0.437	0.01691	0.405	0.472
19	149	4	0.426	0.01744	0.393	0.461
20	130	3	0.416	0.01794	0.382	0.452
21	109	4	0.400	0.01883	0.365	0.439
22	82	4	0.381	0.02029	0.343	0.423
26	48	2	0.365	0.02233	0.324	0.412
27	33	5	0.310	0.02964	0.257	0.374

```
> plot(kmsurvival, xlab="Time", ylab="Survival Probability")
```


Non parametric estimation

Examples with R



Non parametric estimation

Examples with R

```
> # Kaplan-Meier non-parametric analysis by group
> kmsurvival1 <- survfit(Surv(time, event) ~ group)
> summary(kmsurvival1)
Call: survfit(formula = Surv(time, event) ~ group)
```

		group=0						
time	n.risk	n.event	survival	std.err	lower	95 % CI	upper	95 % CI
1	1495	266	0.822	0.00989		0.803		0.842
2	1038	116	0.730	0.01191		0.707		0.754
3	717	55	0.674	0.01317		0.649		0.701
4	501	20	0.647	0.01396		0.620		0.675
5	423	36	0.592	0.01550		0.563		0.623
6	305	8	0.577	0.01603		0.546		0.609
7	265	28	0.516	0.01801		0.482		0.552
8	191	4	0.505	0.01842		0.470		0.542
9	176	5	0.491	0.01898		0.455		0.529
10	151	1	0.487	0.01913		0.451		0.526
11	141	6	0.467	0.02010		0.429		0.508
12	116	1	0.463	0.02033		0.424		0.504
13	111	5	0.442	0.02144		0.402		0.486
14	91	9	0.398	0.02376		0.354		0.447
15	73	3	0.382	0.02459		0.336		0.433
16	61	3	0.363	0.02566		0.316		0.417
17	45	4	0.331	0.02799		0.280		0.390
18	39	3	0.305	0.02944		0.253		0.369
19	35	1	0.297	0.02986		0.243		0.361
26	15	1	0.277	0.03379		0.218		0.352
27	11	1	0.252	0.03897		0.186		0.341

Non parametric estimation

Examples with R

```

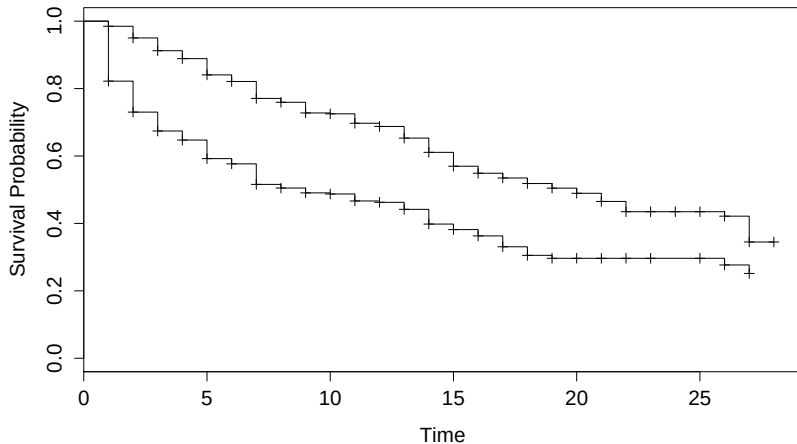
group=1
time n.risk n.event survival std.err lower 95 % CI upper 95 % CI
  1  1848      28   0.985 0.00284   0.979   0.990
  2  1765      62   0.950 0.00511   0.940   0.960
  3  1604      64   0.912 0.00676   0.899   0.926
  4  1396      36   0.889 0.00764   0.874   0.904
  5  1253      68   0.841 0.00919   0.823   0.859
  6  1034      24   0.821 0.00980   0.802   0.841
  7   931      57   0.771 0.01124   0.749   0.793
  8   742      11   0.759 0.01159   0.737   0.782
  9   672      28   0.728 0.01255   0.704   0.753
 10   566       2   0.725 0.01264   0.701   0.750
 11   518      20   0.697 0.01362   0.671   0.724
 12   440       6   0.688 0.01397   0.661   0.716
 13   398      20   0.653 0.01526   0.624   0.684
 14   324      21   0.611 0.01683   0.579   0.645
 15   238      16   0.570 0.01857   0.534   0.607
 16   191       7   0.549 0.01949   0.512   0.588
 17   156       4   0.535 0.02022   0.497   0.576
 18   130       4   0.518 0.02121   0.478   0.562
 19   114       3   0.505 0.02207   0.463   0.550
 20    99       3   0.489 0.02310   0.446   0.537
 21    81       4   0.465 0.02492   0.419   0.517
 22    61       4   0.435 0.02756   0.384   0.492
 26    33       1   0.422 0.02970   0.367   0.484
 27    22       4   0.345 0.04233   0.271   0.439

```

```
> plot(kmsurvival1, xlab="Time", ylab="Survival Probability")
```

Non parametric estimation

Examples with R



Non parametric estimation

Examples with R

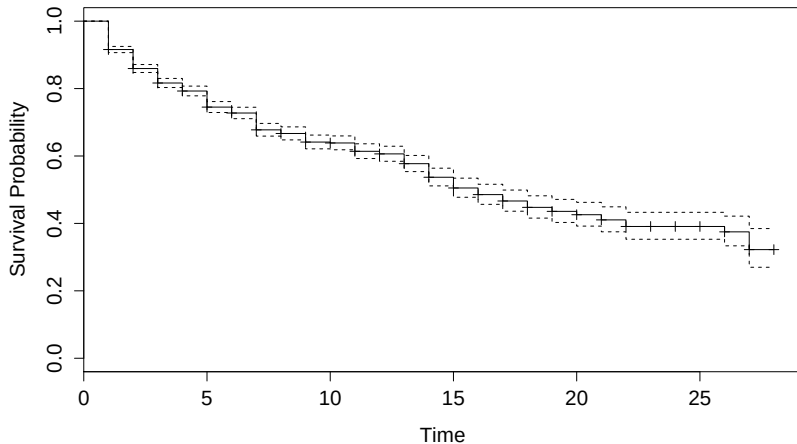
```
> # Nelson-Aalen non-parametric analysis
> nasurvival <- survfit(coxph(Surv(time,event)~1), type="aalen")
> summary(nasurvival)
Call: survfit(formula = coxph(Surv(time, event) ~ 1), type = "aalen")
```

time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
1	3343	294	0.916	0.00470	0.907	0.925
2	2803	178	0.859	0.00601	0.848	0.871
3	2321	119	0.817	0.00688	0.803	0.830
4	1897	56	0.793	0.00738	0.778	0.807
5	1676	104	0.745	0.00828	0.729	0.761
6	1339	32	0.727	0.00865	0.711	0.745
7	1196	85	0.678	0.00960	0.659	0.697
8	933	15	0.667	0.00985	0.648	0.686
9	848	33	0.641	0.01042	0.621	0.662
10	717	3	0.639	0.01049	0.618	0.660
11	659	26	0.614	0.01115	0.592	0.636
12	556	7	0.606	0.01138	0.584	0.629
13	509	25	0.577	0.01223	0.554	0.602
14	415	30	0.537	0.01340	0.511	0.564
15	311	19	0.505	0.01446	0.478	0.534
16	252	10	0.485	0.01517	0.457	0.516
17	201	8	0.467	0.01599	0.436	0.499
18	169	7	0.448	0.01687	0.416	0.482
19	149	4	0.436	0.01743	0.403	0.471
20	130	3	0.426	0.01795	0.392	0.462
21	109	4	0.410	0.01887	0.375	0.449
22	82	4	0.391	0.02035	0.353	0.433
26	48	2	0.375	0.02243	0.333	0.422
27	33	5	0.322	0.02912	0.270	0.385

```
> plot(nasurvival, xlab="Time", ylab="Survival Probability")
```

Non parametric estimation

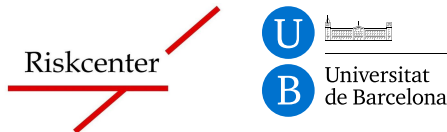
Examples with R



Survival analysis with actuarial applications (I)

Material created by PhD. Ramon Alemany

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MESIO